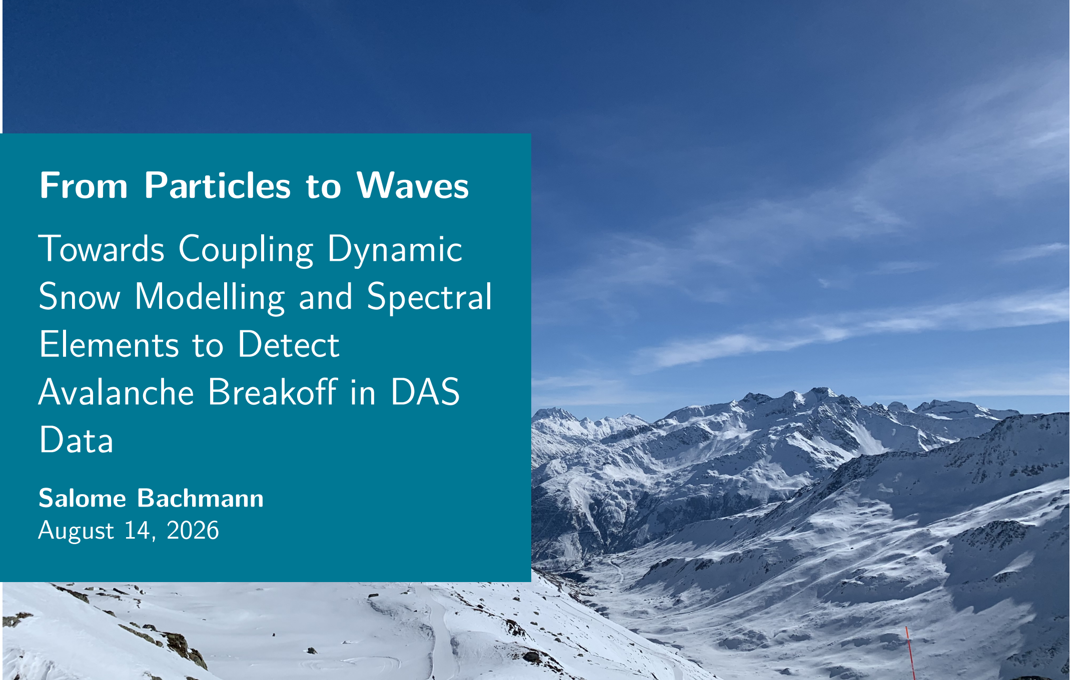


From Particles to Waves

Towards Coupling Dynamic Snow Modelling and Spectral Elements to Detect Avalanche Breakoff in DAS Data

Salome Bachmann

August 14, 2026

Why model crack propagation in snow? - Relevance

- 22 fatalities by avalanche in Switzerland every year (SLF, 2025)
- Avalanches are danger for human life but also infrastructure (Schweizer et al., 2003)
- Understanding processes associated with avalanches important
- Avalanche release because of loading, leading to crack propagation in a weak layer (Gaume et al., 2017)

Relevance

- Crack propagation at different speeds: sub-Rayleigh (below vs of material) and Supershear (above vs) (Trottet et al., 2022), transition between two regimes possible (Bergfeld et al., 2025)
- Fast crack propagation $\rightarrow$ critical crack length for release (Gaume et al., 2015) is reached faster
- Distributed Acoustic Sensing (DAS) already used to monitor avalanches in Switzerland (Edme et al., 2023; Paitz et al., 2023) and Norway (Turquet et al., 2024)
- No knowledge if crack propagation associated with breakoff can be seen in DAS data

Overview

- **Goal**: Couple dynamic snow modelling with SE to model crack propagation in snow
- Created 3 different models and 2 scenarios for crack propagation in snow using SALVUS
- Two pure spectral element (SE) models, one combined with data from material point method (MPM) simulations

Nomenclature & Abbreviations

- **SE, FE, FEM**: Spectral elements, Finite Element (Method), done in SALVUS
- **MPM**: Material Point Method Numerical Modeling
- **PST**: Propagation Saw Test
- **F-k**: Frequency-Wavenumber Plot
- **DAS**: Distributed Acoustic Sensing (Fibre-Optics)
- **STF**: Source-time function (the wavelet fired by SALVUS)

Research Questions

- What FE modelling framework is required to simulate DAS measurements of crack propagation in snow, and what are the key assumptions and limitations?
- What physical phenomena are observed in a two-layer model with a simple STF and how do they differ for sub-Rayleigh versus Supershear crack propagation?
- To what extent do the results change if source wavelets from a MPM simulation of a PST are injected?
- How could the phenomena observed be used to distinguish crack propagation regimes in real-world DAS data and aid in detecting snow slab avalanche breakoff?

Simple Model I

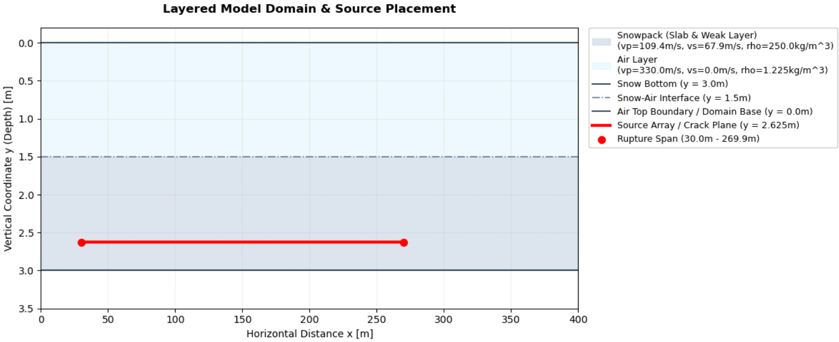


Figure: Two-layer model setup in SALVUS. Colors show materials, material types and properties in legend. Source line extent in red.

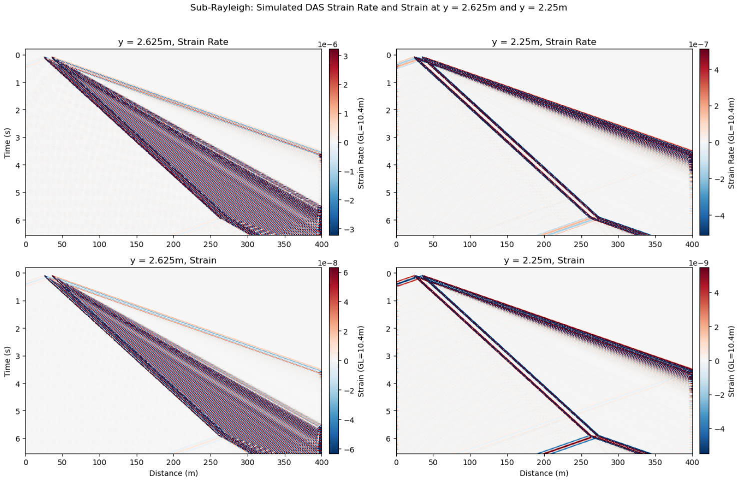
Simple Model II

- Set up to test custom STF in SALVUS, see physical effects on output
- Sources spaced over 240m in 10cm intervals
- 20Hz central frequency Ricker wavelet
- Source Wavelet in xx (1), xy (1) and yy (-1) direction
- Time delay of wavelet calculated based on propagation speed: sub-Rayleigh speed is $0.6v_s$, Supershear $1.6v_s$ (Trottet et al., 2022)

Results

- SALVUS Output: Displacement field u
- $\frac{\partial u_i}{\partial t} \rightarrow$ velocity in x and y
- $\frac{\partial u_i}{\partial j} \rightarrow$ strain in x and y ϵ
- $\frac{\partial^2 u_i}{\partial t \partial t} \rightarrow$ strain rate $\dot{\epsilon}$
- Simulation of DAS data from ϵ, $\dot{\epsilon}$ using Pyber Gauge length class by Noe, 2025, gauge length acoording to Turquet et al., 2024
- Spate-time plots of strain, strain rate and velocity at different reciever depths
- Frequency-wavenumber plots with theoretical speeds and dispersion curves
- Animations of velocity wavefields

Simple Sub-Rayleigh: Strain and Strain Rate I



Simple Sub-Rayleigh: Strain and Strain Rate II

Figure: Horizontal Strain and strain rate for sub-Rayleigh crack propagation in simple two-layer model. Plotted at crack depth (2.625m) and at the snow bottom (2.25m). Gauge length is 10.4m.

- Precursoremitted by all sources but they interfer (Madariaga, 1976), as shown by wider source spacing
- Slope of line close to p-wave speed in medium: S0 wave bounded by plate wave speed limit (“Waves in Plates”, 2014)
- S0/P-wave precursor observed in field data by Herwijnen and Schweizer, 2011
- Oscillatory behavior of both wavefronts (polarity flip at adjacent timesteps)
- Significant weakening of amplitude between two depths

Simple Supershear: Strain and Strain Rate I

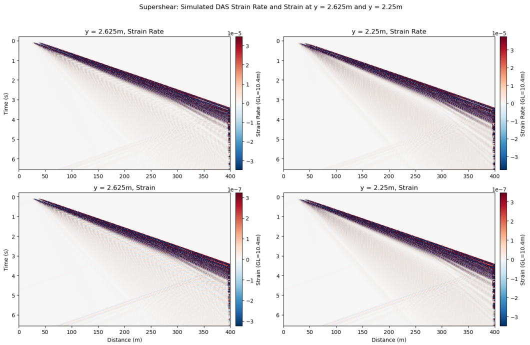


Figure: Horizontal Strain and strain rate for Supershear crack propagation in simple two-layer model. Plotted at crack depth (2.625m) and at the snow bottom (2.25m). Gauge length is 10.4m.

Simple Supershear: Strain and Strain Rate II

- One wavefront, contrary to sub-Rayleigh
- Slope corresponts to rupture speed (within 1m/s of vp and S0)
- Dispersion and broadening of main wavefront (Aki & Richards, 1981)
- Oscillatory wavefront
- At x=270m, unidirectional fan-like arrest front: rupture outruns vs, so arrest energy is only propagated forward (Madariaga, 1977)
- S0 precursor can be seen in traces
- Supershear larger amplitude compared to sub-Rayleigh

Simple model: Lamb Wave Identification I

- Polarity oscillation in all plots: indicative of Lamb waves
- Plotting dispersion curves over f-k spectra: energy is concentrated at symmetric (S0) and anitsymetric (A0) mode
- Antisymmetric mode gets excited preferentially when source is applied normal to midplane of material (Achenbach, 1973; Lamb, 1917; "Waves in Plates", 2014), so in this case yy
- Confirmed by animations of wavefield:

Simple model: Lamb Wave Identification II

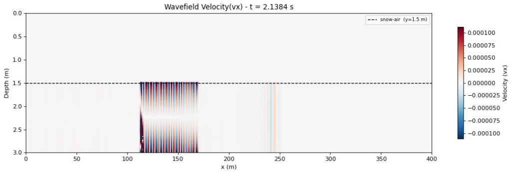


Figure: Animation still for Sub-Rayleigh Vx

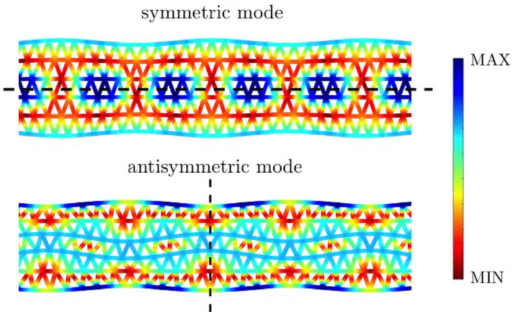


Figure: Lamb wave modes intensity, taken from Carta et al., 2023

MPM-Guided Model I

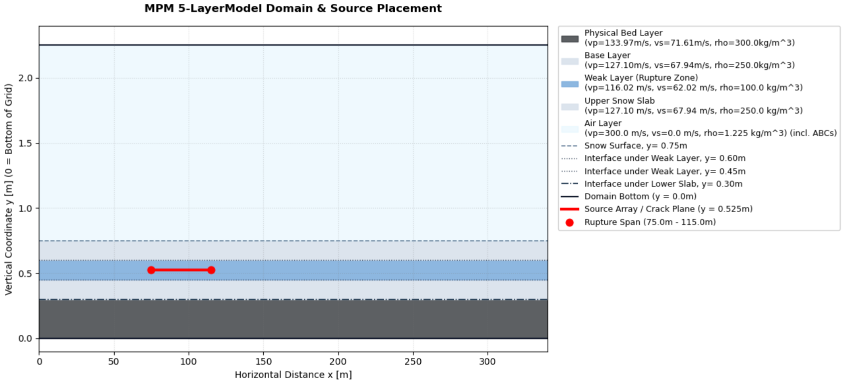


Figure: Five-layer model setup in SALVUS. Colors show materials, material types and properties in legend. Source line extent in red.

MPM-Guided Model II

- Source-time functions obtained from MPM-simulation of PST by Trottet et al., 2022
- Source Wavelet in all three directions, but irregular and different for every source point

MPM-Guided: Strain and Strain Rate I

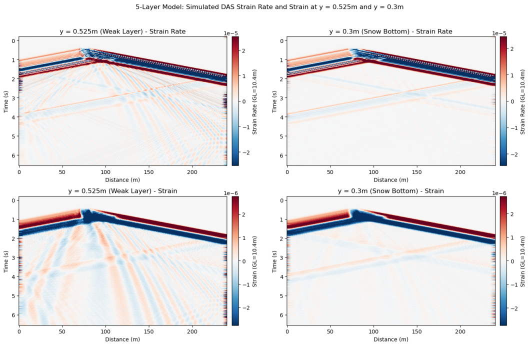


Figure: Horizontal Strain and strain rate for MPM-guided model. Plotted at crack depth (0.525) and at the snow bottom (0.3m). Gauge length is 10.4m.

MPM-Guided: Strain and Strain Rate II

- One two main wavefronts observed
- Crack extent from 75m to 125m: sub-Rayleigh speed from front
- Positive wedge in crack area: extends until rupture arrest, then jumps to same thickness as negative front $\rightarrow$ quasi-static offset
- Backward-propagating front and forward after crack arrest: close to vp/S0 from slope
- Second, weaker wavefront: at onset but also arrest: same arrest structure as sub-Rayleigh
- Oscillatory wavefronts

Quasi-Static Offset

- Static field represents permanent deformation after sources have fired (Archambeau, 1968; Zhu & Rivera, 2002)
- MPM STFS are not symmetric: contrary to Ricker wavelets, they do not integrate to 0 over time $\rightarrow$ net displacement of material (Aki & Richards, 1981; Zhu & Rivera, 2002)
- Cumulative integral of MPM STFs is negative: collapse of the weak layer, as observed by (Gaume et al., 2018; Heierli et al., 2008; Heierli et al., 2003; Trottet et al., 2022)
- All components of MPM STF (directionality) indicate mixed-mode anticrack instead of pure opening or shearing (mode I, II or II)

MPM-Guided Model: Lamb Wave Identification I

- Like in simple model: oscillatory wavefronts → A0 mode

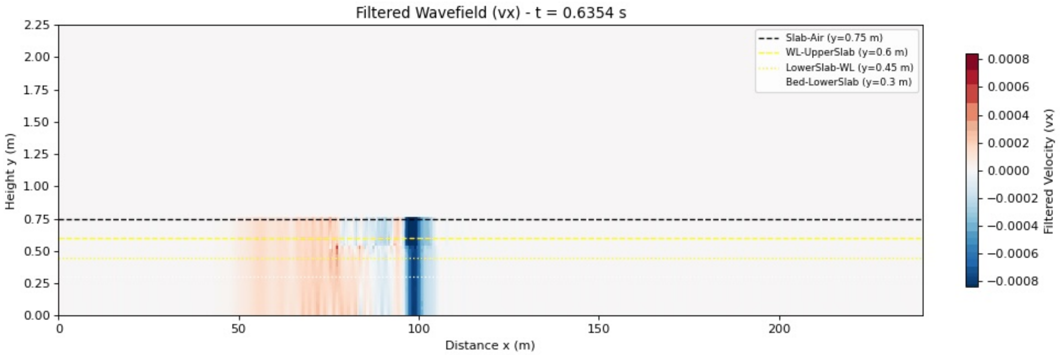


Figure: Animation still for MPM Vx

MPM-Guided Model: Lamb Wave Identification II

- F-k plots also show concentration of energy around A0 and S0 dispersion curves
- Polarity reversal around slab midplane in animation indicative of A0

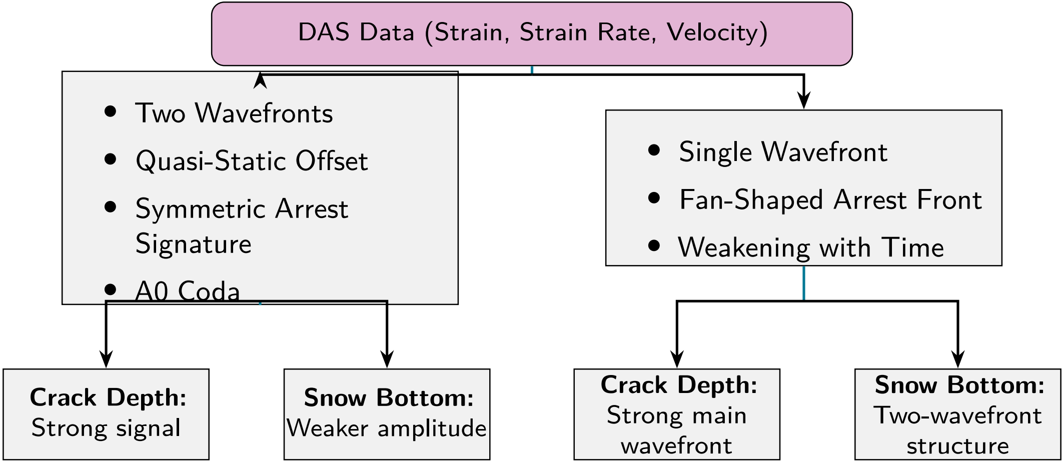
Summary of Results & Discussion I

- S0 Precursor observed in all Strain & Strain Rate Plots, as well as in traces
- Main wavfront in all cases is A0 $\rightarrow$ source normal to slab midplane
- Different rupture arrest structures depending on propagation speed
- Quasi-static wavefield for sources that do not integrate to 0 over time
- Weakening of amplitude only in simple case $\rightarrow$ also thicker slab

Implications for DAS field data I

- Identification of rupture speed via number of wavefronts & arrest structure
- Precursors in sub-Rayleigh: S0 and Quasi-Static offset
- Consistent with observations by Herwijnen and Schweizer, 2011
- Structures identified as being geometry-independent
- Signals identified as being larger than instrument noise of Silixa iDAS used by Paitz et al., 2023
- Supershear signals larger in amplitude: possibly false conclusions if transition happens
- Identification flowchart:

Implications for DAS field data II



Implications for DAS field data III

Figure: Graphical Representation to Identify Different Signatures in DAS

Outlook

- Add topography to models
- Verify Supershear effects for more complex STFs as well
- Inhomogeneous, multi-layered snowpack
- Transitional propagation regimes
- Comparison to field data $\rightarrow$amplitude of signals compared to noise
- 3D model

Answer to Research Questions - Conclusions

- **What physical phenomena are observed in a two-layer model with a simple STF and how do they differ for sub-Rayleigh versus Supershear crack propagation?** Rupture-arrest signatures depending on propagation speed, S0 precursor, A0 main wavefront for both propagation regimes. Changes in amplitude
- **To what extent do the results change if source wavelets from a MPM simulation of a propagation saw test (PST) are injected?** Verification of sub-Rayleigh S0 precursor, A0 mode excitation, changes because appearance of quasi-static offset
- **How could the phenomena observed be used to distinguish crack propagation regimes in real-world DAS data and aid in detecting snow slab avalanche breakoff?** Precursors (S0 and quasi-static offset) used as "early" warning, identification of propagation speed using arrest structures, possibility of identifying transitions by number of wavefronts

Questions?

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MPM STF Workflow

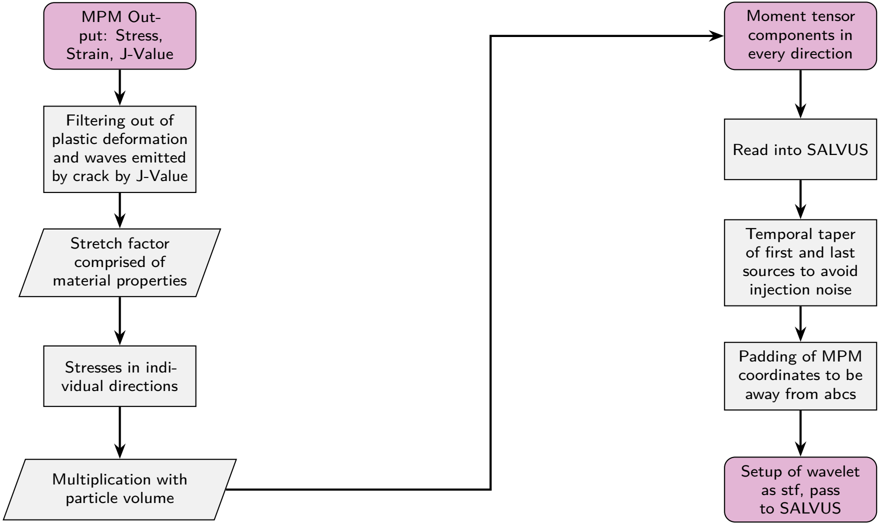


Figure: Workflow for conversion of MPM data and input as SALVUS STF.

Simple model velocity plots I

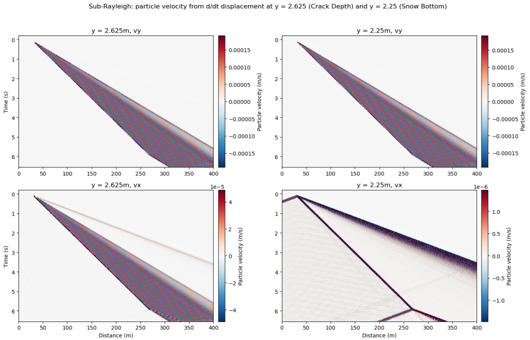


Figure: Simple model Sub-Rayleigh velocity plots. Crack depth on left side, snow bottom on right. Vy at top, vx at bottom. Velocity derived from displacement.

Simple model velocity plots II

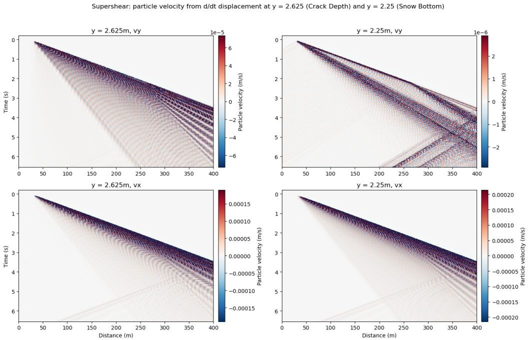


Figure: Simple model Supershear velocity plots. Crack depth on left side, snow bottom on right. Vy at top, vx at bottom. Velocity derived from displacement.

MPM velocity plots

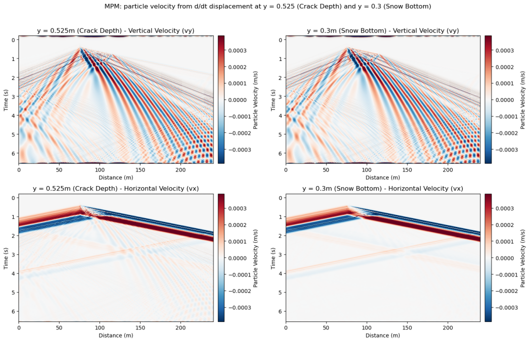
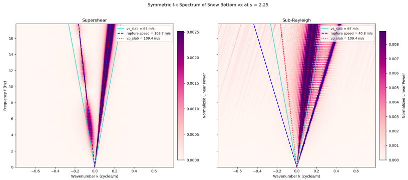


Figure: MPM-guided model velocity plots. Crack depth on left side, snow bottom on right. Vy at top, vx at bottom. Velocity derived from displacement.

Simple model F-k plots and dispersion curves



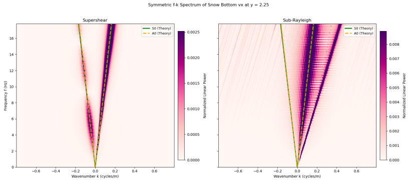
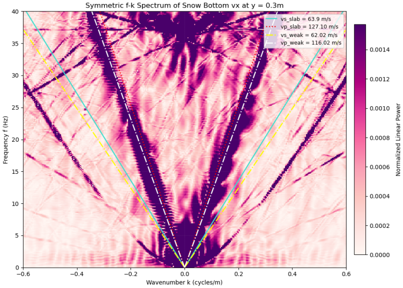


Figure: Supershear and Sub-Rayleigh F-k spectra with medium and Lamb-wave dispersion curves indicated by colored lines.

MPM F-k plots



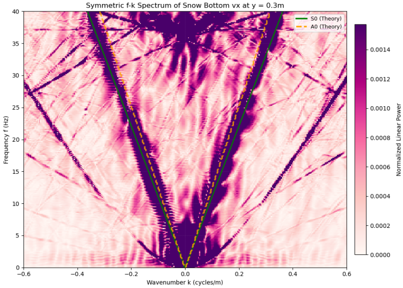


Figure: MPM F-k spectra with medium and Lamb-wave dispersion curves indicated by colored lines.

MPM STF cumulative Integral

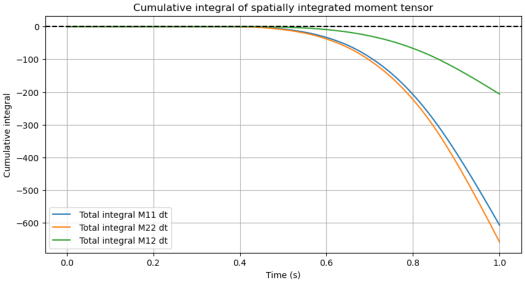


Figure: Cumulative integral over time of all MPM STF components

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